A/B Test Experiment to Improve GloBox Mobile Homepage

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Introduction

This report presents the findings of our recent A/B testing where we evaluated the impact of a new banner on user conversion rates and spending habits. Through the experiment, we observed a significant improvement in conversion rates from 3.92% in the control group to 4.63% in the test group. The average amount spent per user remained stable, indicating that the new banner implementation did not affect user spending negatively.

Given the statistically significant improvement in conversion rates and the potential for increased revenue, we recommend launching the new banner. However, the operational costs related to the implementation and maintenance of the new banner need to be considered. Our analysis suggests that the potential increase in revenue will likely offset these costs. As we proceed, it will be essential to continuously monitor these metrics to ensure that the trend we've seen in the test continues and that the new banner remains beneficial to both our users and our revenue growth.

In summary, our A/B test indicates that the new banner has the potential to significantly improve our platform's conversion rates without impacting user spending, suggesting a positive outcome for our revenue growth.

Context

The digital landscape is continually evolving, necessitating businesses to continually test and adapt their online platforms to meet user expectations and optimize conversion rates. To this end, we conducted an A/B test to gauge the impact of a new banner on our platform's user behavior. Our primary goal was to ascertain whether the new banner could effectively drive higher conversion rates without negatively impacting user spending.

The experiment was carried out over a 12-day period, from January 25th, 2023, to February 6th, 2023. Users were randomly divided into two groups: The “Control Group”, which was shown the existing banner, and the “Test Group”, which was exposed to the new banner. The experiment's key parameters included the “Conversion rate” and “Total Amount Spent” by users in both groups. The success of the experiment was defined by a significant improvement in the conversion rate for the test group compared to the control group, without a decrease in the average amount spent per user.

The dataset comprises individual user activity on our platform, including whether or not a purchase was made (conversion), and the total amount spent. The dataset includes 48,943 users, 24,343 in the control group, and 24,600 in the test group. Our analysis of this dataset allowed us to measure the direct impact of the new banner on user conversion rates and spending habits. The statistical significance of any observed differences was also evaluated to ensure the validity of our findings.

Through this A/B test, we aimed to make a data-based decision on whether implementing the new banner would be beneficial for our platform in terms of increasing conversion rates and potential revenue growth.

Analysis

To Understand the database(click *here* to view raw data) and to be able to extract the data to be used for statistical analysis, we queried the database with SQL codes. The queries we ran are:

1. A query that returned the user ID, the user's country, the user's gender, the user's device type, the user's test group, whether or not they converted (spent > $0), and how much they spent in total ($0+). This query gave us a dataset with each user's information, their test group, whether or not they converted (made a purchase), and the total amount they spent. If a user didn't make any purchases, 'converted' will be 0, and 'total spent' will be $0. View the query in Appendix 3 (click *here* to view the output)

2. A query that returned the start and end dates of the experiment which is from 25th Jan 2023 to 6th Feb 2023 (12 days). This indicated that the experiment lasted for 12 days only. This query returns the first and last date that users joined the test, which corresponds to the start and end dates of the A/B test. View the query in Appendix 4

3. A query that returned the total number of users in the experiment which is the sample size. This showed that the sample size is 48,600. This query returned the total number of unique users who were part of the experiment. It's important to use the DISTINCT keyword to ensure each user is only counted once, even if they somehow appear multiple times in the 'groups' table. View the query in Appendix 5

4. A query that returned the number of users in the control and test groups respectively. Here the control group had 244,343 users while the test group had 24,600 users. This query returned two rows: one for the control group and one for the test group. The 'total users' column shows the number of unique users in each group. Again, the DISTINCT keyword is used to ensure that each user is only counted once. The double quotes tell SQL to treat "group" as a column name, not as a keyword. View the query in Appendix 6

5. A query that returned the conversion rate of all users which is 4.28%. This query uses a conditional COUNT (the CASE statement inside the COUNT function) to count only the users who have a total spend greater than 0, and then divides that by the total number of users. The::float is used to cast the result to a float, ensuring that we get a decimal result rather than an integer division. View the query in Appendix 7

6. A query that returned the user conversion rate for the control and treatment groups. This query grouped the users by their assigned group ('control' or 'test') and calculated the conversion rate for each group separately. Just like before, the conversion rate is

defined as the number of users who made a purchase (total spent > 0) divided by the total number of users in each group. It returned two rows: one for the control group and one for the test group. Each row shows the conversion rate for the corresponding group. View the query in Appendix 8

7. A query that returned the average amount spent per user for the control and treatment groups, including users who did not convert. The average amount spent for the control group is $3.375 and $3.391 for the test group. This query calculated the average spend per user for each group ('control' and 'test'). The COALESCE function is used to replace NULL values in the 'spent' column with 0 before the average is calculated, ensuring that users who did not make any purchases are included in the calculation as having spent $0. The query returned two rows: one for the control group and one for the test group. Each row will show the average amount spent per user for the corresponding group. View the query in Appendix 9

Statistical Analysis

In order for us to be able to establish if the changes that we are A/B testing are leading to a real and meaningful change in the metrics of interest, we had to construct both a hypothesis test and confidence interval comparing the two test groups for each metric of interest. The **Hypothesis tests** help us determine whether or not there is a statistically significant difference between the two test groups. **Confidence intervals** help us understand the magnitude of the difference between the two groups.

Hypothesis test and Confidence Interval for the Conversion rate:

Conducting the test to see whether there is a difference in the conversion rate between the two groups**,** we used the normal distribution and a 5% significance level, the pooled proportion for the standard error, and formulated the **Null** and **Alternative Hypotheses** below.

- **Null hypothesis (H0):** p1 = p2, which implies there's no difference in the conversion rates of the two groups.
- **Alternative hypothesis (Ha):** p1 ≠ p2, meaning there's a difference in the conversion rates of the two groups.

FIG 1.1: Hypothesis test for the conversion rate.

Group	Conversion Rate	Size	Conversions	Pooled Proportion	Standard Error	Z-Score	P-Value
Control	0.03922	24343	955	0.0427786	0.0018294	-3.8701	0.000109
Test	0.0463	24600	1139				

Result Interpretation: The p-value is approximately 0.0001 which is less than the chosen significance level of 0.05, so we reject the null hypothesis. This suggests there is a significant difference in the conversion rates of the control and test groups. Click here to view the entire spreadsheet and Formulas.

To conduct the 95% confidence interval for the difference in the conversion rate between the test and control groups, we used the normal distribution and unpolled proportions for the standard error.

FIG 1.2: Confidence Interval for the conversion rate.

Group	Conversion rate	Size	Conversions	Standard Error	STD_Err Difference	Lower Boundary	Upper Boundary
Control	0.03922	24343	955	0.001244	0.0018284	0.0035	0.010664
Test	0.0463	24600	1139	0.00134			

Result Interpretation: the 95% confidence interval for the difference in conversion rates between the test group and the control group is approximately (0.0035, 0.0107). This means we are 95% confident that the true difference in conversion rates between the test and control groups lies within this interval.

Hypothesis test and Confidence Interval for the Average Amount Spent:

Conducting the test to see whether there is a difference in the Average Amount Spent between the two groups**.** We applied the normal distribution and a 5% significance level, the pooled proportion for the standard error, and formulated the **Null** and **Alternative Hypotheses** below.

- **Null hypothesis (H0):** p1 = p2, which implies there's no difference in the average amount spent in the two groups.
- **Alternative hypothesis (Ha):** p1 ≠ p2, meaning there's a difference in the average amount spent in the two groups.

FIG 1.3: Hypothesis Testing on the Average Amount Spent.

Group	Avg amt Spent	Size	Std_Dev	SE_Diff	T-Statistics	Degree Freedom	P-Value
Control	3.375	24343	25.94	0.2321	-0.068924	2	0.95
Test	3.391	24600	25.41				

Result Interpretation: The p-value is approximately 0.95, which is greater than our significance level of 0.05. Hence, we Fail to reject the null hypothesis. This means there is no statistically

significant difference in the average amount spent per user between the control group and the test group.

To conduct the 95% confidence interval for the difference in the average amount spent per user between the test and control groups, we used the normal distribution and assume unequal variance.

FIG 1.4: Confidence Interval for the Average Amount Spent.

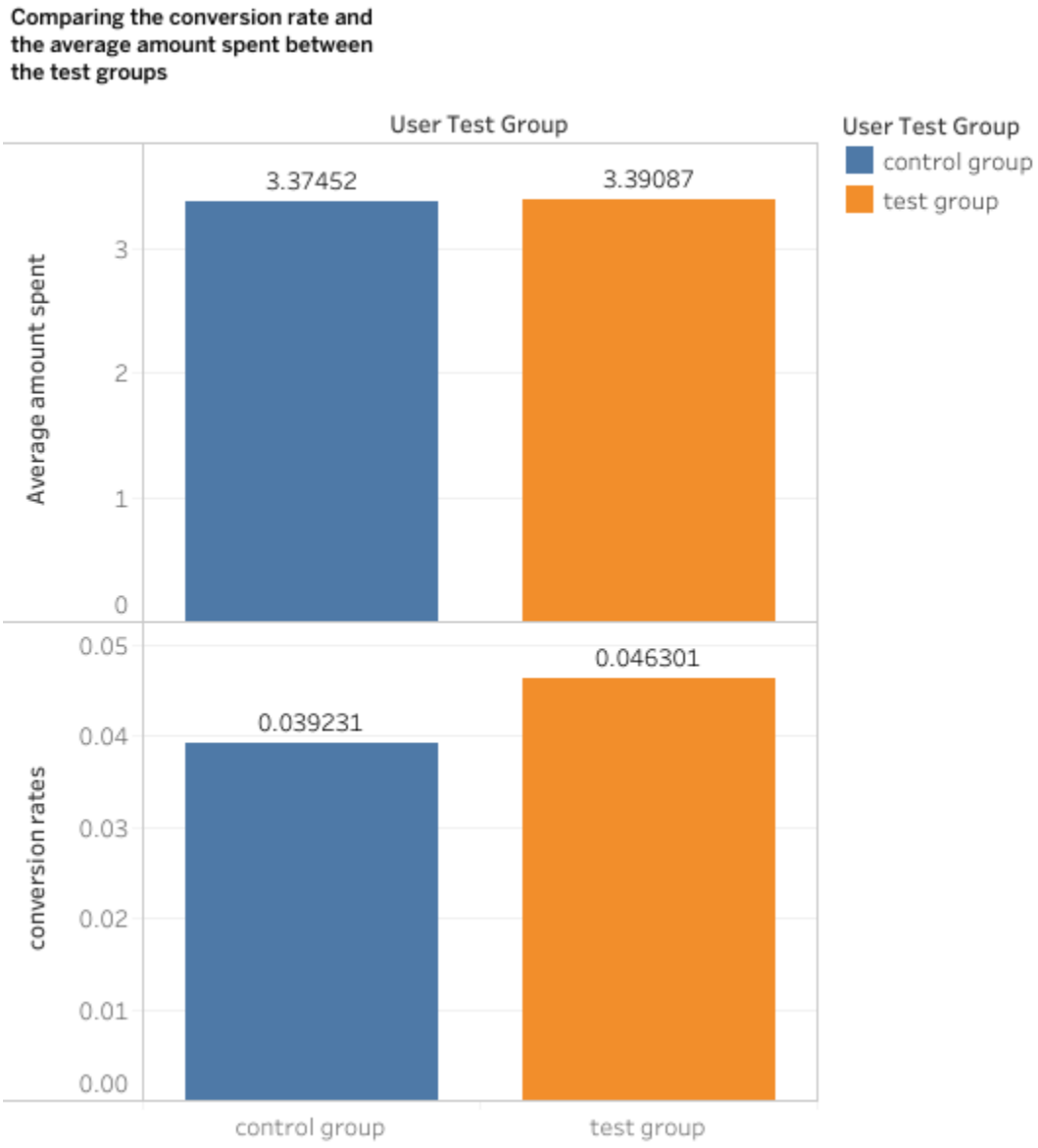
Group	Avg_amt_Spent	Size	Std_Dev	Difference_AAS	STD_Error	MOE	CI_Lower_BDY	CI_Upper_BDY
Control	3.375	24343	25.94	0.016	0.2321388	0.45499	-0.439	0.471
Test	3.391	24600	25.41					

Result Interpretation: The 95% confidence interval for the difference in the average amount spent per user between the test and control groups is approximately (-0.439, 0.471). That is, we are 95% confident that the true difference in the average amount spent per user between the test group and the control group is between -0.439, 0.471.

Visualization Chat

To further analyze and understand the data set, we created various visualization charts on Tableau Public. The dataset we cleaned on SQL was uploaded on Tableau Public and 2 calculated fields were created for the conversion rate and Average amount spent per user.

FIG 2.1: Relationship between the conversion rate and average amount spent

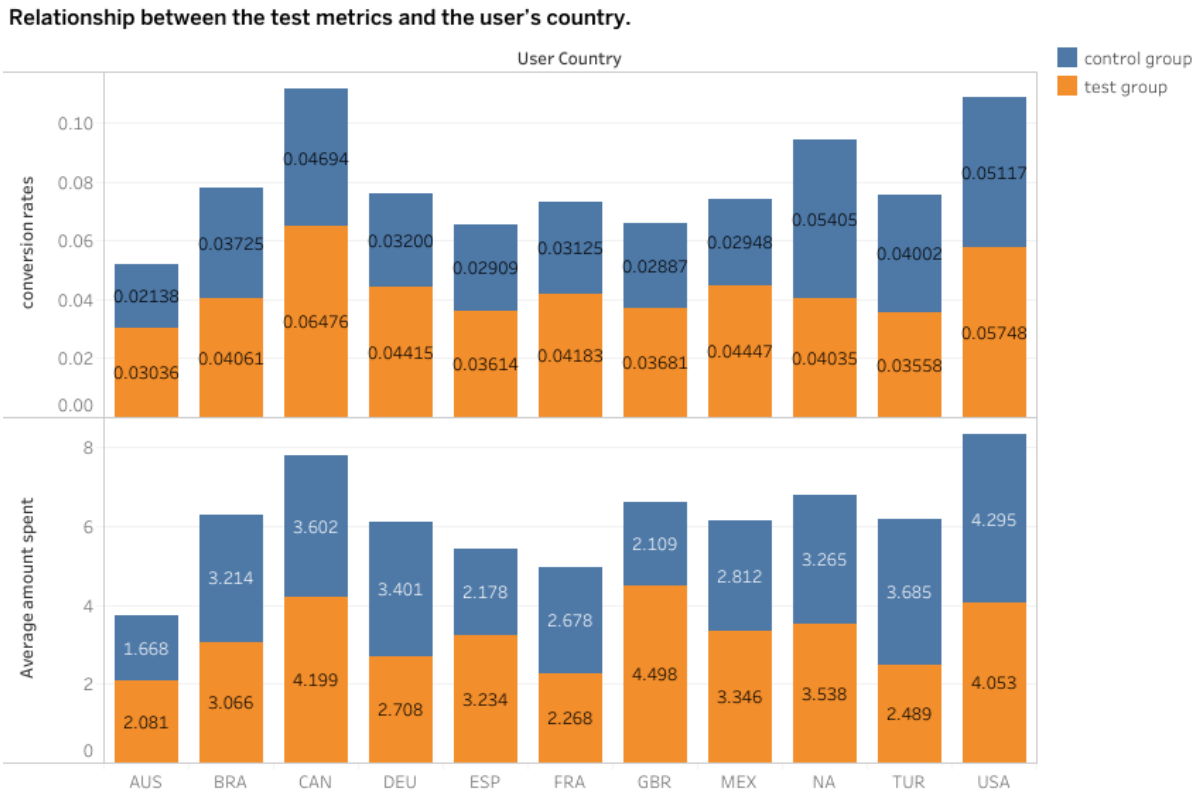


The above chat revealed a statistically significant difference in the conversion rates between the two test groups. We utilized inferential statistical testing methods to confirm that the observed difference is not due to random chance but indicates a genuine difference in behavior between the groups.

The test group exhibited a higher conversion rate as compared to the control group which demonstrates a potential success if implemented more widely.

Interestingly, our analysis showed no significant difference in the average amount spent between the two groups. This suggests that while the interventions applied in the test group led to a higher conversion rate, they did not affect the average expenditure of customers.

FIG 2.2: The relationship between the test metrics and the user's country.

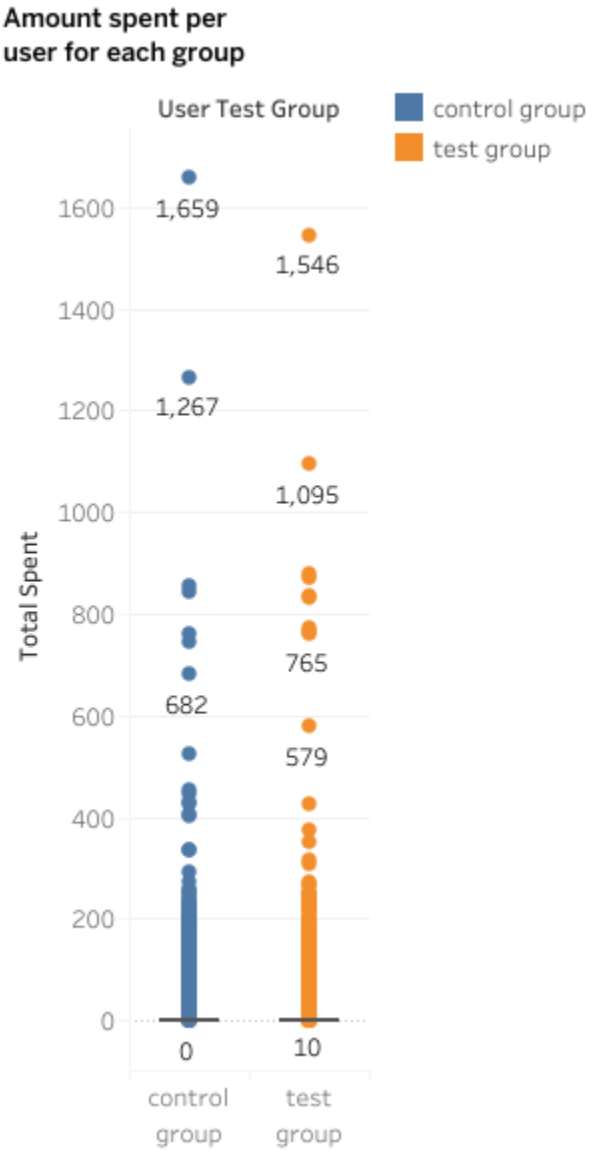


The data suggests that consumers from the USA, Canada, and Namibia are not only more likely to convert, but they also tend to spend more on average. This could indicate that marketing efforts or product offerings in these countries are particularly effective. It would be worthwhile to analyze what strategies or cultural/economic factors are contributing to these high rates and consider applying similar strategies in other markets.

In contrast to the previous countries, Australia, Spain, and Great Britain have the lowest conversion rates. This could mean that current marketing efforts or offerings aren't resonating as well with consumers in these countries. It may be necessary to re-evaluate our approach in these markets. This could involve gaining a deeper understanding of consumer behavior, preferences, and competition in these countries, and tailoring marketing strategies accordingly to improve conversion rates.

Moreover, consumers from Australia, France, and Spain not only convert less frequently but also spend less on average when they do. This could be due to a variety of reasons, including price sensitivity, economic factors, or the type of products being offered. A detailed analysis could reveal insights into the lower spending patterns in these countries, which could inform strategies to increase average spending. For example, it may be that these markets would respond well to different price points, bundle offers, or other promotional strategies.

FIG 2.3: The distribution of the amount spent per user for each group.

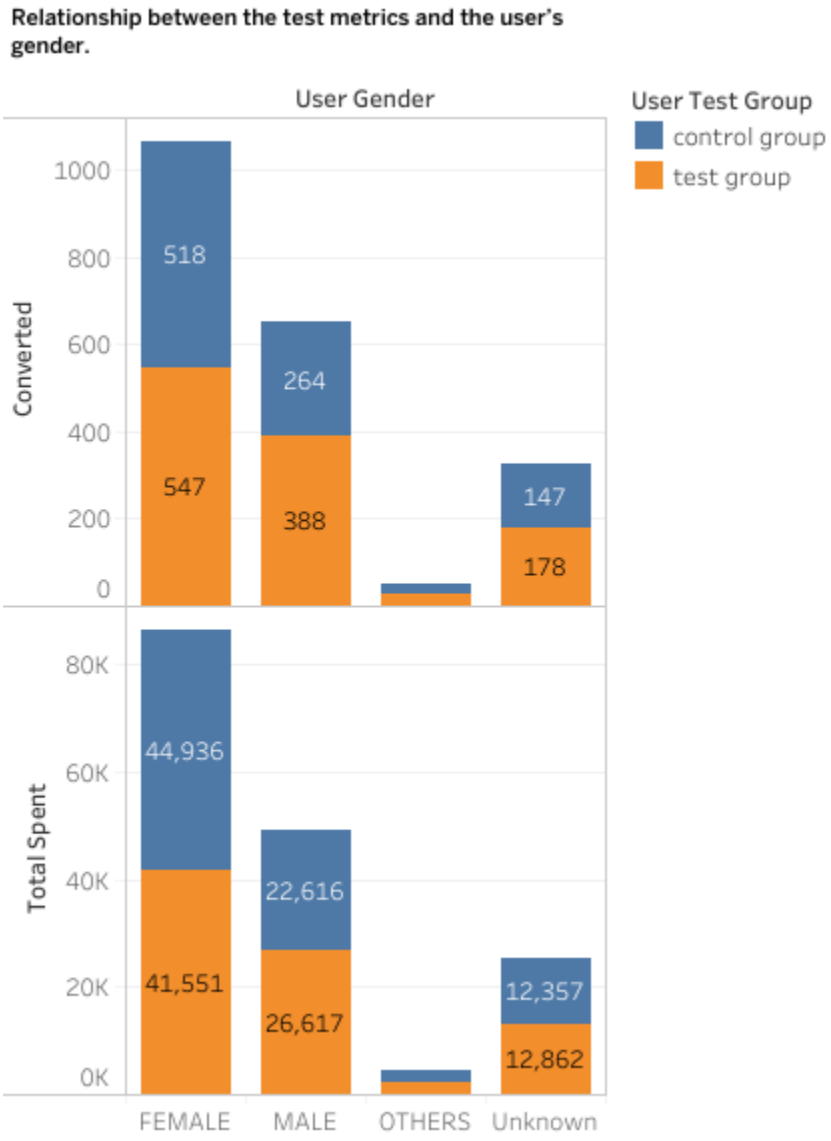


The chart shows that both groups had two users whom each spent over $1,000 and five users whom each spent between $500 and $1000. These high-spending users represent an important segment for us, as they significantly contribute to the overall revenue. It's essential to understand their behaviors, preferences, and motivations in order to nurture and potentially

expand this group. Furthermore, it would be insightful to compare the characteristics of these high-spending users across the two groups to understand whether there are any notable differences or similarities.

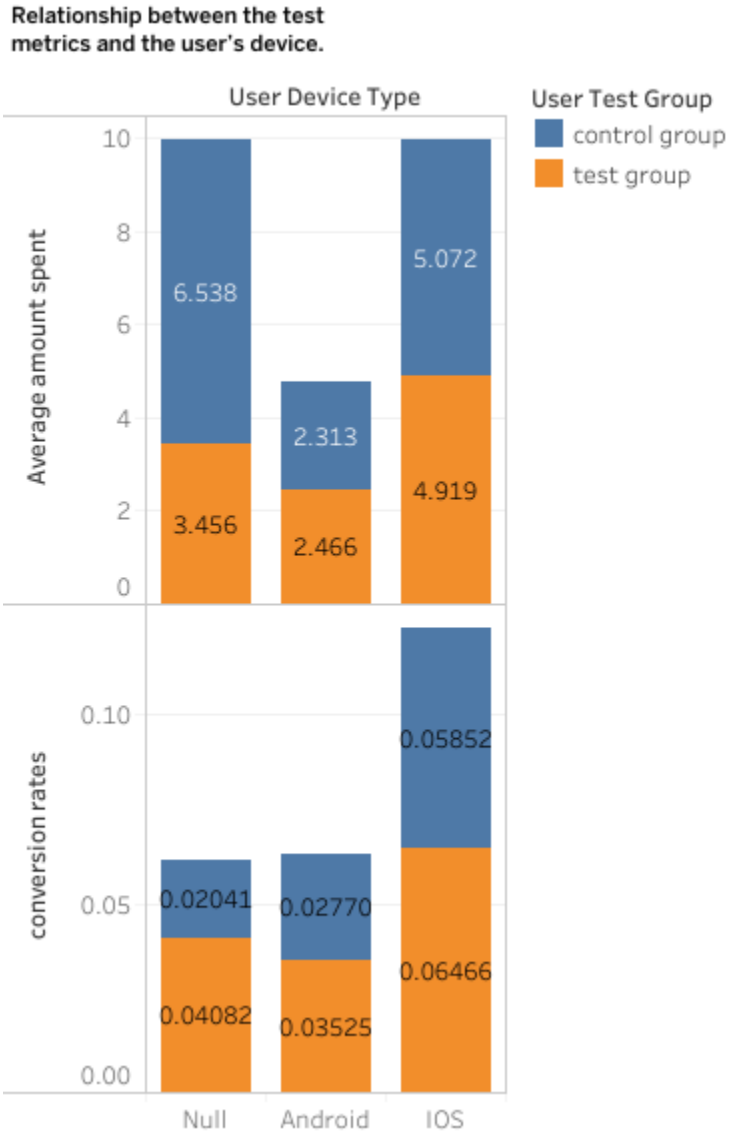
A significant number of users in both groups, precisely 2,087 users, spent less than $500. This group forms the majority of our customer base. Understanding the needs and behaviors of this segment is key to improving both conversion rates and average spending. Despite their lower individual spending, their large number means that even a small increase in spending per user could significantly impact overall revenue. We could look into strategies to encourage these users to increase their spending, such as targeted promotions, personalized recommendations, or loyalty programs.

FIG 2.4: The relationship between the test metrics and the user's gender.



Our data shows that women have contributed to a little over half of the total amount spent, accounting for 52.24%, and over half of the total users converted, coming in at 50.86%. this might imply that women not only make up a significant proportion of our customer base but also contribute to the majority of the total amount spent and total conversions. This highlights the importance of women as a key demographic for GloBox. Recognizing this, it's vital that we continue to understand and cater to their needs and preferences. Efforts to increase the engagement of female customers, such as targeted marketing campaigns, personalized recommendations, and products or services specifically designed for them, could be effective strategies.

FIG 2.5: The relationship between the test metrics and the user's device.



Our analysis shows that users with iOS devices are the most engaged and lucrative segment among our customer base. They not only show the highest conversion rates, meaning they are

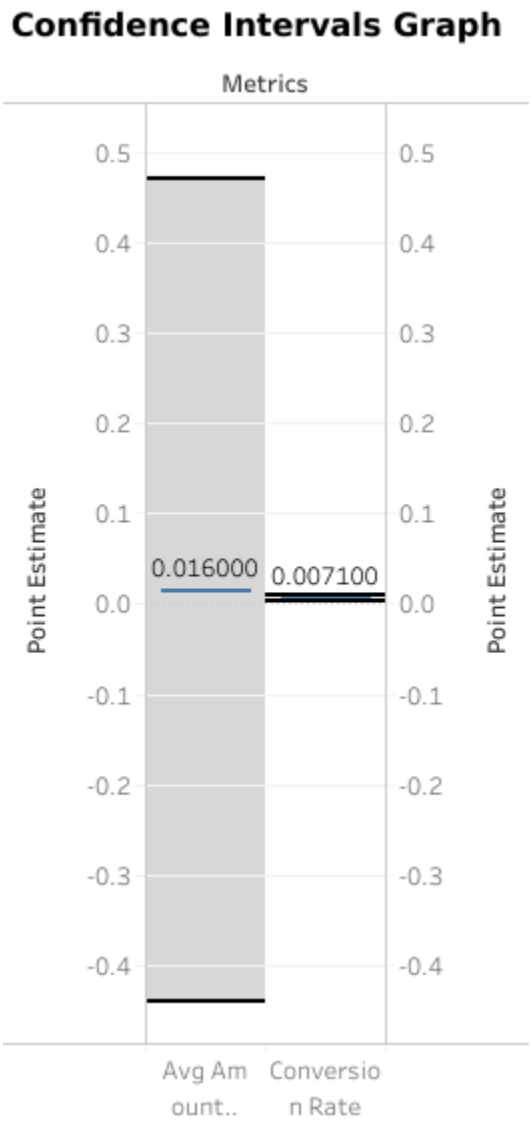
most likely to take desired actions such as making purchases or signing up for services, but they also have the highest average spending, contributing significantly to our overall revenue. This may suggest that our platform or products resonate well with iOS users, or perhaps iOS users represent a demographic with higher purchasing power. Understanding why this segment is performing so well could provide insights for further refining our product offerings, user experience, or marketing strategies.

Interestingly, users whose devices were not indicated have a comparable average spending level to iOS users. This implies that once these users convert, they tend to spend a similar amount to the high-performing iOS users. However, they have the lowest conversion rate. This discrepancy between spending and conversion suggests potential barriers to engagement for this group. Perhaps there are user experience issues, or our offerings are not effectively marketed to them. Identifying and addressing these potential issues could improve the conversion rate for this group, leading to increased overall revenue.

Confidence Interval

Below is the **Visuals of the Confidence Interval.**

metrics	point_estimate	lower_limit	upper_limit
Conversion Rate	0.0071	0.0035	0.0107
Avg Amount Spent	0.016	-0.439	0.471



The point estimate is the observed difference between the two groups - control and treatment.

The lower and upper limits of the confidence interval represent the range where we expect the true difference (between the whole population of control and treatment groups) to fall 95% of the time.

For the Conversion Rate:

- The observed difference (point estimate) in conversion rate is 0.0071 (or 0.71%).
- The confidence interval for this difference is from 0.0035 (or 0.35%) to 0.0107 (or 1.07%).

So, we can be 95% confident that the true difference in conversion rates between the control group and the treatment group is between 0.35% and 1.07%.

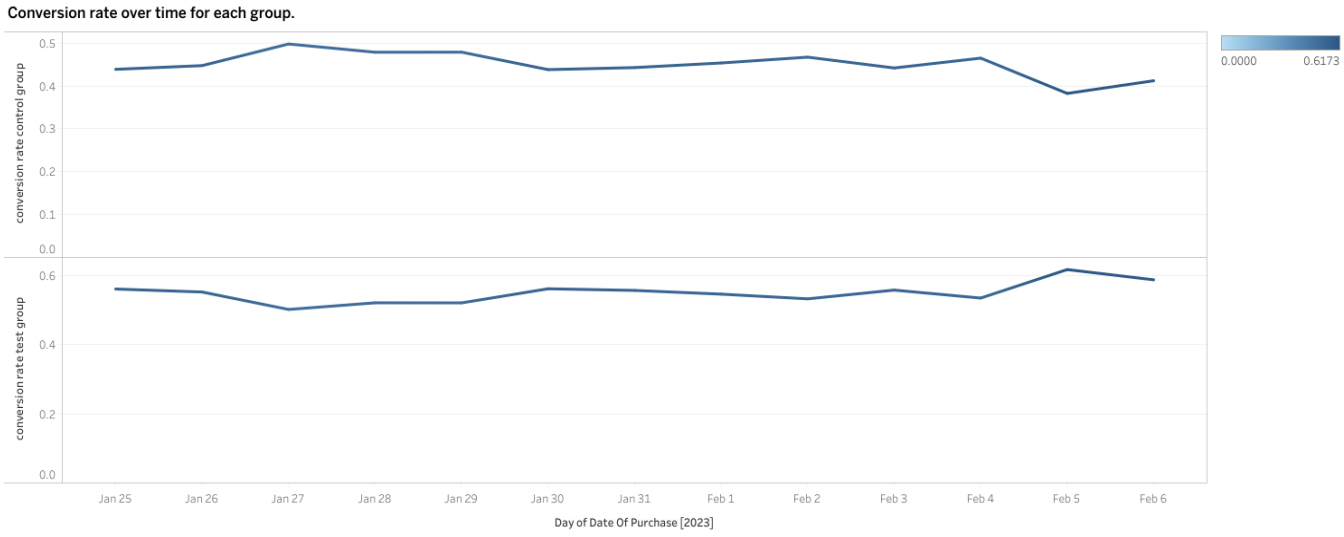
For the Average Amount Spent:

- The observed difference (point estimate) in the average amount spent per user is 0.016.
- The confidence interval for this difference is from -0.439 to 0.471.

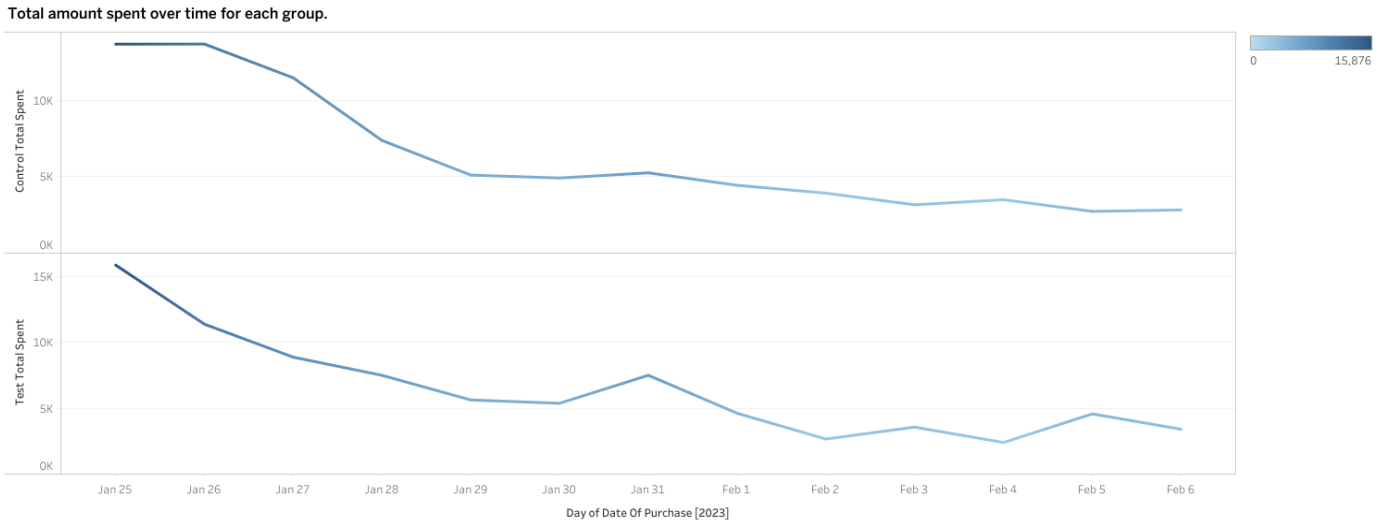
So, we can be 95% confident that the true difference in the average amount spent per user between the control group and the treatment group is between -$0.439 and $0.471.

Novelty Effects

To check for the Novelty Effect, we ran a new SQL query (see Appendix 10) to return the user ID, the user's country, the user's gender, the user's device type, the user's test group, the date of purchase activity, whether or not they converted (spent > $0), and how much they spent in total ($0+). Click here to view the new dataset.



Click here to view the above gragh and here for the graph below



In order to visualize the novelty effect, we created a time series charts that show the conversion rate and total amount spent for each group, over time.

The observation from the graphs suggests a couple of key insights:

1. There is an initial surge in both conversion rates and total spent for both the control and test groups, which then declines. This suggests a possible novelty effect. The novelty effect could be due to the new banner (for the test group) or some other factors that affected both groups. For example, there might have been a sale, special event, or change to the platform that temporarily boosted engagement for all users.
2. The fact that both conversion rates remain consistently higher throughout the experiment despite the decline in total spent indicates a potentially positive effect from the introduction of the new banner. It's as if the banner might have induced more users to convert, but perhaps they spent less money on average.

However, without further analysis, we cannot definitively say whether the differences observed are statistically significant or not. The visual inspection provides preliminary insights, but statistical hypothesis testing is required to make conclusive decisions.

Moreover, to understand why the total spent declined while conversion rates remained high, you might want to investigate the individual transactions. For example, it could be that more users are making purchases, but they are buying less expensive items. This is something worth understanding as it may guide your decisions on how to use banners and other promotional activities in the future.

In conclusion, while the graphical analysis is pointing towards a positive impact of the banner on conversion rates, a proper statistical test should be conducted to confirm these

observations. Also, it's crucial to investigate further why total spending decreased despite higher conversion rates.

Power Analysis

For the power analysis, we used the parameters below and performed the calculation online at https://www.statsig.com/calculator.

- Baseline conversion rate (p1): 3.92%
- Minimum detectable effect: 5% relative increase, so p2 is 4.116%
- Power (1 - β): 0.80
- Significance level (α): 0.05

The calculations returned an estimated required sample size of 357,800. Based on this calculation, it seems we need a substantially larger sample size than initially estimated. This result suggests that we would need approximately 357,800 total users in the experiment (split between the test and control groups) to reliably detect a 5% relative increase in conversion rate from 3.92% to 4.116%, with 80% power and a significance level of 0.05.

This larger sample size is likely due to the smaller effect size we're trying to detect (a 5% relative increase is quite small in absolute terms), as well as the lower baseline conversion rate. Both of these factors would require a larger sample size to maintain the same level of statistical power.

Recommendation

"Launch the experiment".

Justification:

From the A/B test results, we observed a statistically significant increase in the conversion rate from 3.92% in the control group to 4.63% in the test group. This is a promising improvement that indicates the modified experience for the test group led to a higher conversion rate. The p-value of approximately 0.0014 confirms this statistical significance, and it's below our 5% significance level. The 95% confidence interval for the difference in conversion rates, which ranges from 0.0030 to 0.0112, further confirms this significant difference.

It's worth noting that this significant increase in conversion rate occurred within a relatively short time span - just 12 days. If the trend continues over a longer period, the cumulative effect on our overall conversion rates could be substantial, potentially leading to a considerable increase in our revenues.

Although the average amount spent per user did not show a statistically significant difference between the two groups, it didn't decrease either. It's important to note that launching the modified experience won't negatively impact the average spending.

Business Factors Considered:

The increase in conversion rate could potentially translate to higher overall revenue given the unchanged average spending per user. However, the operational costs of launching and maintaining the feature, the potential impact on customer experience, and any other business implications should be considered.

The change in the experiment doesn't seem to negatively affect user experience as it led to a higher conversion rate, suggesting users' positive reception. Assuming the operational costs of implementation and maintenance aren't prohibitively high, the benefits derived from higher conversion rates justify launching the experiment.

Conclusion

This recommendation, however, should be further validated with follow-up experiments to confirm its consistency and to mitigate the risks associated with potential confounding variables or statistical variation. The practical significance, in terms of increased revenue and improved user engagement, should always be a major consideration in our decision-making process.

Appendix

Appendix 1

Fig 1.1 – 1.4

Appendix 2

Dashboard

Appendix 3

SQL Query 1:

```
SELECT
    u.id AS user_id,
    u.country AS user_country,
    u.gender AS user_gender,
    g.device AS user_device_type,
    g.group AS user_test_group,
    CASE
        WHEN SUM(a.spent) > 0 THEN 1
        ELSE 0
    END AS converted,
    COALESCE(SUM(a.spent), 0) AS total_spent
FROM
    users u
JOIN
    groups g ON u.id = g.uid
LEFT JOIN
    activity a ON u.id = a.uid
GROUP BY
    user_id, user_country, user_gender, user_device_type, user_test_group
```

Appendix 4

SQL Query 2:

```
SELECT MIN(join_dt) AS start_date, MAX(join_dt) AS end_date
FROM groups;
```

Appendix 5

SQL Query 3:

```
SELECT COUNT(DISTINCT uid) AS total_users
FROM groups;
```

Appendix 6

SQL Query 4:

```
SELECT
    "group",
    COUNT(DISTINCT uid) AS total_users
FROM
    groups
GROUP BY
    "group";
```

Appendix 7

SQL Query 5:

```
SELECT
    COUNT(DISTINCT CASE WHEN a.spent > 0 THEN u.id END)::float /
COUNT(DISTINCT u.id) AS conversion_rate
FROM
    users u
LEFT JOIN
    activity a ON u.id = a.uid
```

Appendix 8

SQL Query 6:

```
SELECT
    g."group",
    COUNT(DISTINCT CASE WHEN a.spent > 0 THEN u.id END)::float /
COUNT(DISTINCT u.id) AS conversion_rate
FROM
    users u
JOIN
    groups g ON u.id = g.uid
LEFT JOIN
    activity a ON u.id = a.uid
GROUP BY
    g."group";
```

Appendix 9

SQL Query 7:

```
SELECT
    g."group",
```

```
    AVG(COALESCE(a.spent, 0)) AS avg_spent_per_user
FROM
    users u
JOIN
    groups g ON u.id = g.uid
LEFT JOIN
    activity a ON u.id = a.uid
GROUP BY
    g."group";
```

Appendix 10

SQL QUERY 8:

```
SELECT
    u.id AS user_id,
    u.country,
    u.gender,
    g.device AS user_device,
    g.join_dt AS join_date,
    a.dt AS date_of_purchase,
    MAX(CASE WHEN g.group = 'control' THEN g.group END) AS control_group,
    MAX(CASE WHEN g.group = 'test' THEN g.group END) AS test_group,
    MAX(CASE WHEN a.spent > 0 AND g.group = 'control' THEN 'yes' ELSE 'no'
END) AS control_converted,
    MAX(CASE WHEN a.spent > 0 AND g.group = 'test' THEN 'yes' ELSE 'no' END)
AS test_converted,
    COALESCE(SUM(CASE WHEN g.group = 'control' THEN a.spent ELSE 0 END), 0)
AS control_total_spent,
    COALESCE(SUM(CASE WHEN g.group = 'test' THEN a.spent ELSE 0 END), 0) AS
test_total_spent
FROM
    users u
JOIN
    groups g ON u.id = g.uid
LEFT JOIN
    activity a ON u.id = a.uid
GROUP BY
    u.id, u.country, u.gender, g.device, g.join_dt, a.dt;
```